

PERAM JOTHIRNADH REDDY

Chennai, Tamil Nadu, India

Phone: +91 9515096362

Email: p.jothirnadhreddy@gmail.com

LinkedIn: <https://www.linkedin.com/in/jothirnadhreddy-peram-204025311/>

GitHub: <https://github.com/JOTHIRNADHREDDY>

PROFESSIONAL SUMMARY

I am a Mechatronics Engineering student at S.R.M Institute of Science and Technology. I have a strong interest in robotics, automation, artificial intelligence, and entrepreneurship. I enjoy combining mechanical design, electronics, and software to create smart systems that tackle real-world problems. I view engineering as a way to turn ideas into practical and scalable solutions. I focus on learning through hands-on projects, solving problems, and growing my technical skills. My main interests include embedded systems, computer vision, control systems, and intelligent mechanical design. I also have experience developing autonomous and semi-autonomous robots that integrate hardware, software, and real-time control.

EDUCATION

Bachelor of Technology in Mechatronics Engineering

SRM Institute of Science and Technology (SRM IST), Chennai

August 2024 – May 2028 (Expected)

CGPA: 9.51

Intermediate Education

Sri Chaitanya College of Education

Board of Intermediate Education Andhra Pradesh

Jun 2022 – Mar 2024

Grade: 92.2%

Secondary Education

Sri Chaitanya High School

Board of Secondary Education Andhra Pradesh

Jun 2021 – Mar 2022

Grade: 93.5%

CORE TECHNICAL AREAS

Embedded Systems | Computer Vision | Control Systems | Pneumatic Systems |
Autonomous Robotics | ESP32 Development | Mechatronic System Design

TECHNICAL PROJECTS

Semi-Autonomous Weed Detection and Laser Removal Robot

Technologies: YOLOv11s, ESP32 DevKit, OpenCV, Arduino IDE, SolidWorks
Feb 2026 – Mar 2026

- Developed a semi-autonomous precision agriculture robot capable of detecting and targeting weeds using computer vision and laser-based actuation.
 - Extracted pixel coordinates from detected bounding boxes and applied calibration-based coordinate transformation to map image coordinates into real-world targeting positions.
 - Designed a galvanometer-inspired servo laser positioning mechanism for accurate targeting without repositioning the chassis.
 - Integrated BTS7960 motor driver and 12V DC geared motors for stable locomotion.
 - Combined AI inference, embedded control, and mechanical actuation into a unified system for sustainable farming applications.
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Air-Powered Pneumatic Safety Bumper

Technologies: SolidWorks, Pneumatic Double-Acting Cylinders
September 2025 – December 2025

- Designed and developed an automatic pneumatic bumper mechanism to reduce vehicle collision impact using compressed air technology.
 - Modeled the complete mechanical assembly and pneumatic circuit system in SolidWorks.
 - Utilized double-acting pneumatic cylinders to extend the front bumper pre-impact for kinetic energy absorption..
 - Applied principles of fluid mechanics, energy absorption, and mechanical design optimization to improve system efficiency and reliability.
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IoT-Based Trash Collecting and Segregation Robot

Technologies: Arduino, ESP32, Blynk, Metal Detection Module

Apr 2025 – May 2025

- Developed a manually controlled waste collection robot with wireless IoT-based operation.
 - Programmed microcontroller to control robotic arm, DC motors, and metal detection-based segregation mechanism.
 - Enabled remote control via Blynk platform to reduce direct human interaction with hazardous waste.
 - Designed mechanical chassis and storage compartment for efficient waste handling.
 - Collaborated within a 3-member team on electronics, embedded logic, and structural integration.
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Self-Balancing Robot

Technologies: Arduino Uno, MPU6050 IMU, C++, PID Control

Nov 2024 – Dec 2024

- Designed and built a two-wheeled self-balancing robot implementing a closed-loop PID control system.
 - Processed accelerometer and gyroscope data from MPU6050 for tilt estimation.
 - Implemented real-time PID algorithm in C/C++ to maintain vertical stability.
 - Tuned proportional, integral, and derivative gains experimentally to reduce oscillations and improve settling response.
 - Gained practical experience in feedback systems, sensor fusion fundamentals, and motor control.
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CERTIFICATIONS & ACHIEVEMENTS

Awarded Best Hardware Design

Under the theme *Agricultural Robotics* at the **National Level Automation & Robotics Challenge (ARC'26)** organized by SRM Institute of Science and Technology

100 Days of Code: The Complete Python Pro Bootcamp

Udemy | Issued Mar 2025

TECHNICAL SKILLS

Programming:

C | Python

Embedded & Electronics:

ESP32 | Arduino (Uno, Nano) | Sensor Interfacing | PWM Motor Control | Motor Drivers (BTS7960) | Basic PCB Concepts

Computer Vision & AI:

Basics YOLO-based Object Detection Concepts

Control Systems:

PID Controllers

Mechanical & Pneumatics:

Pneumatic Systems | Mechanical Assembly Design

CAD & Simulation:

SolidWorks | LTSpice

Tools:

Arduino IDE | VS Code

RESEARCH & TECHNICAL INTERESTS

Autonomous Systems | Space Robotics | Embedded AI | Vision-Guided Control | Sustainable Robotics Solutions

LANGUAGES

English (Fluent) | Telugu (Native) | Hindi (Conversational)